```c
#include <stdio.h>
#include <string.h>

struct Product {
    int id;
    char name[50];
    int quantity;
    float price;
};

void addProduct()
{
    FILE *fp = fopen("stock.txt", "a");
    struct Product p;

    printf("Enter Product ID: ");
    scanf("%d", &p.id);
    printf("Enter Product Name: ");
    scanf("%s", p.name);
    printf("Enter Quantity: ");
    scanf("%d", &p.quantity);
    printf("Enter Price: ");
    scanf("%f", &p.price);

    fprintf(fp, "%d %s %d %.2f\n", p.id, p.name, p.quantity, p.price);
    fclose(fp);

    printf("Product '%s' added successfully!\n", p.name);
}

void displayProduct()
{
    FILE *fp = fopen("stock.txt", "r");
    struct Product p;
    int lowstock=0;

    printf("\n STOCK LIST\n");
    printf("ID Name Quantity Price\n");

    while (fscanf(fp, "%d %s %d %f", &p.id, p.name, &p.quantity, &p.price) != EOF) {
        if (p.quantity < 5) {
            printf("%d %s %d %.2f\n", p.id, p.name, p.quantity, p.price);
            lowstock=1;
        } else {
            printf("  %d %s %d %.2f\n", p.id, p.name, p.quantity, p.price);
        }
    }

    if (lowstock=1) {
        printf("ALERT: Some products have LOW STOCK!\n");
    }

    fclose(fp);
}

void searchProduct()
```

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{
    FILE *fp = fopen("stock.txt", "r");
    struct Product p;
    int id, found = 0;

    printf("Enter Product ID to search: ");
    scanf("%d", &id);

    while (fscanf(fp, "%d %s %d %f", &p.id, p.name, &p.quantity, &p.price) != EOF) {
        if (p.id == id)
            {
            printf("\nProduct Found:\n");
            printf("ID: %d\n", p.id);
            printf("Name: %s\n", p.name);
            printf("Quantity: %d", p.quantity);
            if (p.quantity < 5) printf("(LOW STOCK!)");
            printf("\nPrice: %.2f\n", p.price);
            found = 1;
            break;
        }
    }

    if (found=0) {
        printf("Product ID %d not found.\n", id);
    }

    fclose(fp);
}
void modifyProduct() {
    FILE *fp = fopen("stock.txt", "r");
    FILE *temp = fopen("temp.txt", "w");
    struct Product p;
    int id, found = 0;

    printf("Enter Product ID to modify: ");
    scanf("%d", &id);

    while (fscanf(fp, "%d %s %d %f", &p.id, p.name, &p.quantity, &p.price) != EOF) {
        if (p.id == id) {
            printf("Current details:\n");
            printf("Name: %s, Qty: %d, Price: %.2f\n", p.name, p.quantity, p.price);

            printf("Enter new Product Name: ");
            scanf("%s", p.name);
            printf("Enter new Quantity: ");
            scanf("%d", &p.quantity);
            printf("Enter new Price: ");
            scanf("%f", &p.price);
            found = 1;
        }
        fprintf(temp, "%d %s %d %.2f\n", p.id, p.name, p.quantity, p.price);
    }

    fclose(fp);
    fclose(temp);
    remove("stock.txt");
    rename("temp.txt", "stock.txt");
```

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    if (found)
        printf("Product updated successfully!\n");
    else
        printf("Product not found.\n");
}
void deleteProduct() {
    FILE *fp = fopen("stock.txt", "r");
    FILE *temp = fopen("temp.txt", "w");
    struct Product p;
    int id, found = 0;

    printf("Enter Product ID to delete: ");
    scanf("%d", &id);

    while (fscanf(fp, "%d %s %d %f", &p.id, p.name, &p.quantity, &p.price) != EOF) {
        if (p.id == id) {
            printf("Deleting: %s (Qty: %d)\n", p.name, p.quantity);
            found = 1;
            continue;
        }
        fprintf(temp, "%d %s %d %.2f\n", p.id, p.name, p.quantity, p.price);
    }

    fclose(fp);
    fclose(temp);
    remove("stock.txt");
    rename("temp.txt", "stock.txt");

    if (found!=0)
        printf("Product deleted successfully!\n");
    else
        printf("Product not found.\n");
}
void checkLowStock() {
    FILE *fp = fopen("stock.txt", "r");
    struct Product p;
    int lowstock=0;

    printf("\n LOW STOCK CHECK\n");

    while (fscanf(fp, "%d %s %d %f", &p.id, p.name, &p.quantity, &p.price) != EOF) {
        if (p.quantity < 5) {
            printf("%s (ID:%d): %d units left | Price: %.2f\n",
                   p.name, p.id, p.quantity, p.price);
            lowstock=1;
        }
    }

    if (lowstock=0) {
        printf(" All products have good stock levels!\n");
    } else {
        printf("\n ACTION NEEDED: Restock low stock items!\n");
    }

    fclose(fp);
}
```

```
int main() {
    int op;

    while (1) {
        printf("\n===== STOCK MANAGEMENT SYSTEM =====\n");
        printf("1. Add Product\n");
        printf("2. Display All Products\n");
        printf("3. Search Product\n");
        printf("4. Modify Product\n");
        printf("5. Delete Product\n");
        printf("6. Check Low Stock\n");
        printf("Enter your choice: ");
        scanf("%d", &op);

        switch (op) {
            case 1: addProduct();
            break;
            case 2: displayProduct();
            break;
            case 3: searchProduct();
            break;
            case 4: modifyProduct();
            break;
            case 5: deleteProduct();
            break;
            case 6: checkLowStock();
            break;
            default: printf("Invalid choice! Try again.\n");
        }
    }

    return 0;
}
```